

Clancey et al. MechanisticModel (STOPPP)

SIR Model with Waning Antibodies and Waning Immunity

N is the total population size, S is the number of susceptible individuals, I is the number of infected individuals, R_A is the number of seropositive and recovered individuals, and R_T is the number of seronegative and recovered individuals.

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In[1]:=  $\dot{N} = \dot{S} + \dot{I} + \dot{R}_A + \dot{R}_T$ 
Out[1]=  $\dot{R}_A + \dot{R}_T + \dot{S} + \dot{I}$ 

In[2]:=  $\dot{S} = b N - \beta S I - S (\mu + k N) + \omega_T R_T$ 
Out[2]=  $-S \beta I + b N - S (\mu + k N) + R_T \omega_T$ 

In[3]:=  $\dot{I} = \beta S I - \gamma I - I (\mu + k N)$ 
Out[3]=  $S \beta I - \gamma I - I (\mu + k N)$ 

In[4]:=  $\dot{R}_A = \gamma I - R_A (\mu + k N) - \omega_A R_A$ 
Out[4]=  $\gamma I - R_A (\mu + k N) - R_A \omega_A$ 

In[5]:=  $\dot{R}_T = \omega_A R_A - R_T (\mu + k N) - \omega_T R_T$ 
Out[5]=  $-R_T (\mu + k N) + R_A \omega_A - R_T \omega_T$ 

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Equilibria and $\mathcal{R}_0$

Disease Free Equilibria (DFE)

```

In[6]:=  $\dot{N} = \text{FullSimplify}[\dot{S} + \dot{I} + \dot{R}_A + \dot{R}_T] /. \{R_A + R_T + S + I \rightarrow N\}$ 
Out[6]=  $b N - N (\mu + k N)$ 

In[7]:=  $\text{Solve}[\dot{N} == 0, N]$ 
Out[7]=  $\left\{ \{N \rightarrow 0\}, \left\{ N \rightarrow \frac{b - \mu}{k} \right\} \right\}$ 

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In[8]:= Solve[ (S /. {E -> 0, I -> 0, RA -> 0, RT -> 0}) == 0, S] /. {N -> (b - μ)/k}
```

```
Out[8]:= {{S -> (b - μ)/k}}
```

```
In[9]:= DFE = {S -> (b - μ)/k, E -> 0, I -> 0, RA -> 0, RT -> 0};
```

Pathogen Present Equilibria

```
In[10]:= FullSimplify[Solve[{0 == S, 0 == I, 0 == RA, 0 == RT}, {S, I, RA, RT}] /. {N -> (b - μ)/k}]
```

```
Out[10]=
```

$$\left\{ \left\{ S \rightarrow \frac{b - \mu}{k}, I \rightarrow 0, RA \rightarrow 0, RT \rightarrow 0 \right\}, \left\{ S \rightarrow \frac{b + \gamma}{\beta}, I \rightarrow -\frac{(b(k - \beta) + k\gamma + \beta\mu)(b + \omega_A)(b + \omega_T)}{k\beta((b + \gamma)(b + \omega_T) + \omega_A(b + \gamma + \omega_T))}, \right. \right. \\ \left. \left. RA \rightarrow -\frac{\gamma(b(k - \beta) + k\gamma + \beta\mu)(b + \omega_T)}{k\beta((b + \gamma)(b + \omega_T) + \omega_A(b + \gamma + \omega_T))}, RT \rightarrow -\frac{\gamma(b(k - \beta) + k\gamma + \beta\mu)\omega_A}{k\beta((b + \gamma)(b + \omega_T) + \omega_A(b + \gamma + \omega_T))} \right\} \right\}$$

Calculate $\mathcal{R}_0$ using the Next Generation Method (NGM)

Build vectors $\mathcal{F}$ and $\mathcal{V}$ to linearize the *infection subsystem*

$\mathcal{F}$ represents the rate of appearance of new infections in compartment i , $\mathcal{V}_i^+(x)$ represents the rate of transfer of individuals into compartment i by all other means, and $\mathcal{V}_i^-(x)$ represents the rate of transfer of individuals out of compartment i , where $\mathcal{V}_i(x) = \mathcal{V}_i^-(x) - \mathcal{V}_i^+(x)$. The infected compartments $x = \{I\}$ and the non-infected compartments are $y = \{S, RA, RT\}$. Thus, this can be written as $\frac{dx}{dt} = \mathcal{F}(x) - \mathcal{V}(x)$.

```
In[11]:= F = {β S I}
```

```
Out[11]=
```

$$\{S \beta I\}$$

```
In[12]:= V = -{-γ I - I (μ + k N)}
```

```
Out[12]=
```

$$\{\gamma I + I(\mu + k N)\}$$

Find the Jacobian of $\mathcal{F}$ and $\mathcal{V}$ around the DFE

```
In[13]:= F = Grad[F, {I}]
```

```
Out[13]=
```

$$\{S \beta\}$$

In[14]:= **Fdfe = F /. DFE**

Out[14]=

$$\left\{ \left\{ \frac{\beta (b - \mu)}{k} \right\} \right\}$$

In[15]:= **MatrixForm[Fdfe]**

Out[15]//MatrixForm=

$$\left(\frac{\beta (b - \mu)}{k} \right)$$

In[16]:= **V = Grad[V, {I}]**

Out[16]=

$$\{ \{ \gamma + \mu + k N \} \}$$

In[17]:= **Vdfe = V /. DFE**

Out[17]=

$$\{ \{ \gamma + \mu + k N \} \}$$

In[18]:= **MatrixForm[Vdfe]**

Out[18]//MatrixForm=

$$(\gamma + \mu + k N)$$

Take the inverse of $\mathcal{V}$

In[19]:= **Vinverse = Inverse[Vdfe]**

Out[19]=

$$\left\{ \left\{ \frac{1}{\gamma + \mu + k N} \right\} \right\}$$

In[20]:= **MatrixForm[Vinverse]**

Out[20]//MatrixForm=

$$\left(\frac{1}{\gamma + \mu + k N} \right)$$

In[21]:= **FVinverse = Fdfe.Vinverse**

Out[21]=

$$\left\{ \left\{ \frac{\beta (b - \mu)}{k (\gamma + \mu + k N)} \right\} \right\}$$

In[22]:= **MatrixForm[FVinverse]**

Out[22]//MatrixForm=

$$\left(\frac{\beta (b - \mu)}{k (\gamma + \mu + k N)} \right)$$

Find the dominant eigen value of $\mathcal{F} \mathcal{V}^{-1}$ to find $\mathcal{R}_0$

In[23]:= **FullSimplify[Eigenvalues[FVinverse]]**

Out[23]=

$$\left\{ \frac{\beta (b - \mu)}{k (\gamma + \mu + k N)} \right\}$$

$$\text{In[24]:= } R_0 = \frac{\beta (b - \mu)}{k (b + \gamma)}$$

$$\text{Out[24]= } \frac{\beta (b - \mu)}{k (b + \gamma)}$$

Solution for I as a Function of R_A

Solution for I for the Mechanistic Model in Counts

$$\hat{I}(t) = \frac{R_A (\mu + k N + \omega_A) + \dot{R}_A}{\gamma}$$

$$\text{In[25]:= } \text{bfunc} = g \text{ Exp}[-s \text{ Cos}[\pi f t - \psi]^2]$$

$$\text{Out[25]= } e^{-s \text{ Cos}[f \pi t - \psi]^2} g$$

$$\text{In[26]:= } \text{bPars} = \{g \rightarrow 0.004441452, s \rightarrow 4, f \rightarrow 1/365, \psi \rightarrow 1\}$$

$$\text{Out[26]= } \left\{g \rightarrow 0.00444145, s \rightarrow 4, f \rightarrow \frac{1}{365}, \psi \rightarrow 1\right\}$$

$$\text{In[27]:= } \text{bave} = \text{N[Integrate[bfunc /. bPars, \{t, 0, 365\}]]} / 365$$

$$\text{Out[27]= } 0.00137022$$

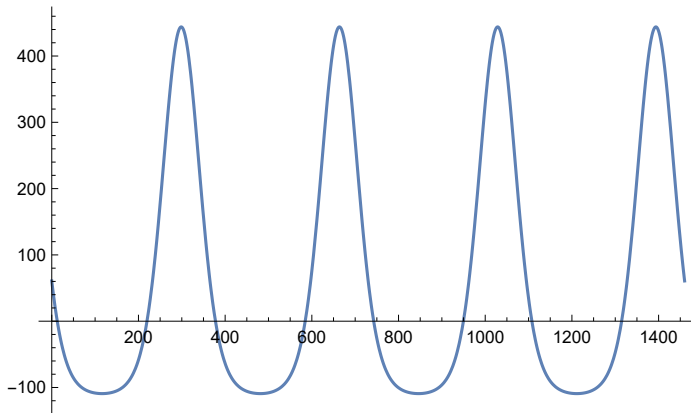
$$\text{In[28]:= } \text{plottime} = 365 * 4$$

$$\text{Out[28]= } 1460$$

$$\text{In[29]:= } \text{Pars} = \{b \rightarrow \text{bfunc} /. \text{bPars}, \beta \rightarrow .0006, \gamma \rightarrow 1/10, \\ \mu \rightarrow 1/1095, k \rightarrow (1/730 - 1/1095) / 10000, \omega_A \rightarrow 1/30, \omega_T \rightarrow 1/180\};$$

In[30]:= **R0Plot = Plot[{R0 /. Pars}, {t, 0, plottime}]**

Out[30]=



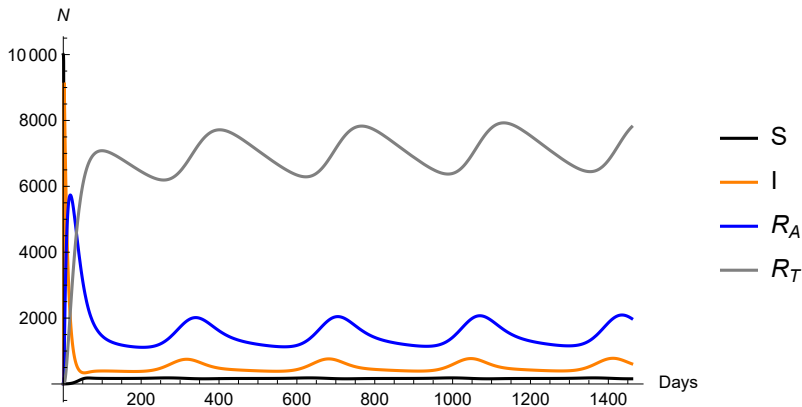
In[31]:= **Sol1 = NDSolve[{S'[t] == b (S[t] + I[t] + RA[t] + RT[t]) -
 $\beta S[t] \times I[t] - S[t] (\mu + k (S[t] + I[t] + RA[t] + RT[t])) + \omega_T RT[t],$
 $I'[t] == \beta S[t] \times I[t] - \gamma I[t] - I[t] (\mu + k (S[t] + I[t] + RA[t] + RT[t])),$
 $RA'[t] == \gamma I[t] - RA[t] (\mu + k (S[t] + I[t] + RA[t] + RT[t])) - \omega_A RA[t],$
 $RT'[t] == \omega_A RA[t] - RT[t] (\mu + k (S[t] + I[t] + RA[t] + RT[t])) - \omega_T RT[t],$
 $S[0] == 9999, I[0] == 1, RA[0] == 0, RT[0] == 0} /. Pars,$
{S[t], I[t], RA[t], RT[t]}, {t, 0, plottime}]**

Out[31]=

$\{ \{ S[t] \rightarrow \text{InterpolatingFunction} \left[\begin{array}{c} \text{Domain: } \{ \{0., 1.46 \times 10^3\} \} \\ \text{Output: scalar} \end{array} \right] [t],$
 $I[t] \rightarrow \text{InterpolatingFunction} \left[\begin{array}{c} \text{Domain: } \{ \{0., 1.46 \times 10^3\} \} \\ \text{Output: scalar} \end{array} \right] [t],$
 $RA[t] \rightarrow \text{InterpolatingFunction} \left[\begin{array}{c} \text{Domain: } \{ \{0., 1.46 \times 10^3\} \} \\ \text{Output: scalar} \end{array} \right] [t],$
 $RT[t] \rightarrow \text{InterpolatingFunction} \left[\begin{array}{c} \text{Domain: } \{ \{0., 1.46 \times 10^3\} \} \\ \text{Output: scalar} \end{array} \right] [t] \} \}$

```
In[32]:= Plot[{S[t] /. Sol1, I[t] /. Sol1, RA[t] /. Sol1, RT[t] /. Sol1},
  {t, 0, plottime}, PlotLegends → {"S", "I", "RA", "RT"},
  PlotStyle → {Black, Orange, Blue, Gray}, AxesLabel → {Days, N}]
```

Out[32]=



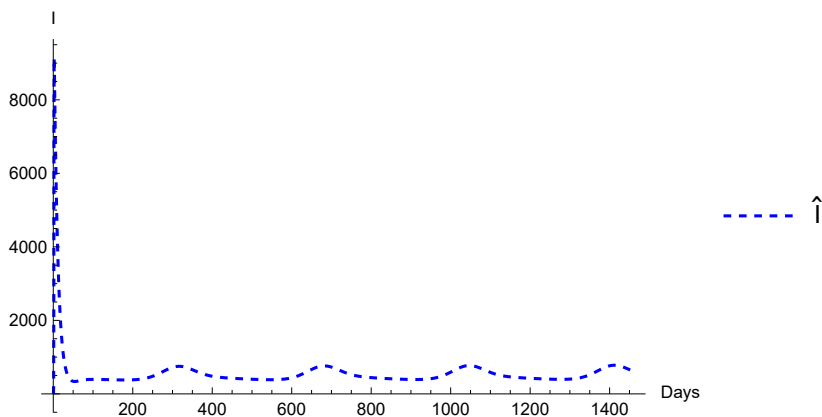
```
In[33]:= IPred =  $\frac{1}{\gamma}$  (D[RA[t] /. Sol1, t] +
  ( $\omega_A + \mu + k$  ((S[t] /. Sol1) + (I[t] /. Sol1) + (RA[t] /. Sol1) + (RT[t] /. Sol1)))
  RA[t] /. Sol1) /. Pars
```

Out[33]=

$$\left\{ \left\{ 10 \left(\text{InterpolatingFunction} \left[\left\{ \left\{ 0., 1.46 \times 10^3 \right\} \right\} \right] [t] \right. \right. \right. \\ \left. \left(\frac{5}{146} + \frac{1}{21900000} \left(\text{InterpolatingFunction} \left[\left\{ \left\{ 0., 1.46 \times 10^3 \right\} \right\} \right] [t] + \right. \right. \right. \right. \\ \text{InterpolatingFunction} \left[\left\{ \left\{ 0., 1.46 \times 10^3 \right\} \right\} \right] [t] + \\ \text{InterpolatingFunction} \left[\left\{ \left\{ 0., 1.46 \times 10^3 \right\} \right\} \right] [t] + \\ \left. \left. \left. \text{InterpolatingFunction} \left[\left\{ \left\{ 0., 1.46 \times 10^3 \right\} \right\} \right] [t] \right) \right) \right) + \right. \\ \left. \left. \left. \text{InterpolatingFunction} \left[\left\{ \left\{ 0., 1.46 \times 10^3 \right\} \right\} \right] [t] \right) \right) \right\} \right\}$$

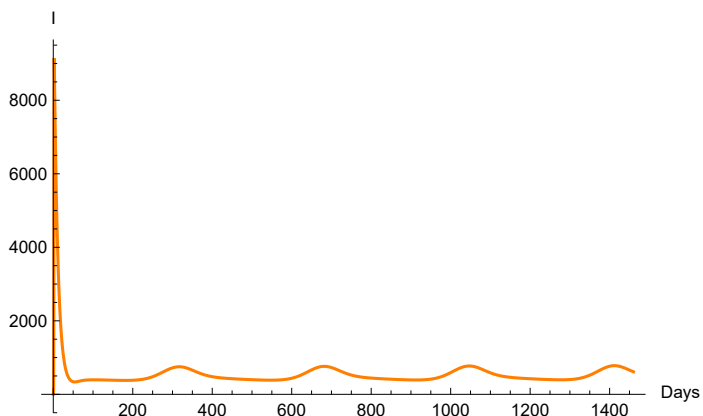
```
In[34]:= Pred = Plot[{IPred /. Sol1}, {t, 0, plottime}, PlotStyle -> {Blue, Dashed},
  PlotRange -> All, AxesLabel -> {Days, I}, PlotLegends -> {" $\hat{I}$ "}]
```

Out[34]=



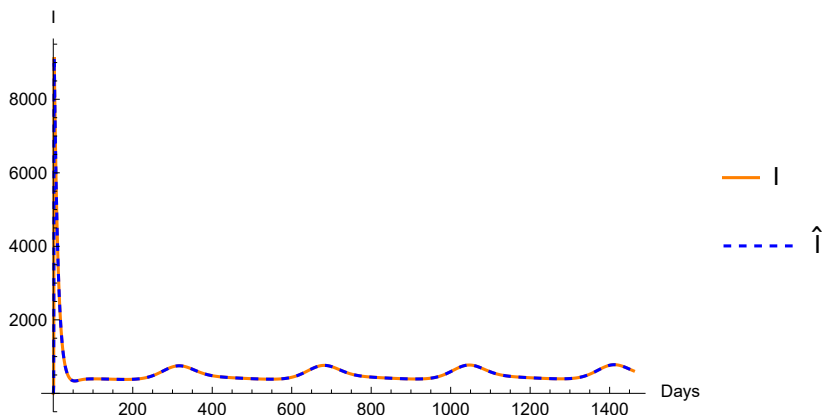
```
In[35]:= TRUE = Plot[{I[t] /. Sol1}, {t, 0, plottime}, PlotRange -> All,
  PlotStyle -> {Orange}, AxesLabel -> {Days, I}, PlotLegends -> {"I"}]
```

Out[35]=



```
In[36]:= Show[TRUE, Pred]
```

Out[36]=



Solution for / for the Mechanistic Model in Proportions

N is the total population size, s is the number of susceptible individuals, i is the number of infected individuals, r_A is the number of seropositive and recovered individuals, and r_T is the number of seronegative and recovered individuals.

Change of Variables (COV) from Counts to Proportions using the quotient rule

$$\text{In[37]:= } \dot{s} = \text{FullSimplify}\left[\text{FullSimplify}\left[\frac{\dot{S} N - \dot{N} S}{N^2}\right] /. \{S \rightarrow s N, I \rightarrow i N, RA \rightarrow r_A N, RT \rightarrow r_T N\}\right]$$

Out[37]=

$$b - b s - s \beta i N + r_T \omega_T$$

$$\text{In[38]:= } \dot{i} = \text{FullSimplify}\left[\text{FullSimplify}\left[\frac{\dot{I} N - \dot{N} I}{N^2}\right] /. \{S \rightarrow s N, I \rightarrow i N, RA \rightarrow r_A N, RT \rightarrow r_T N\}\right]$$

Out[38]=

$$-i (b + \gamma - s \beta N)$$

$$\text{In[39]:= } \dot{r}_A = \text{FullSimplify}\left[\text{FullSimplify}\left[\frac{\dot{R}_A N - \dot{N} R_A}{N^2}\right] /. \{S \rightarrow s N, I \rightarrow i N, RA \rightarrow r_A N, RT \rightarrow r_T N\}\right]$$

Out[39]=

$$-b r_A + \gamma i - r_A \omega_A$$

$$\text{In[40]:= } \dot{r}_T = \text{FullSimplify}\left[\text{FullSimplify}\left[\frac{\dot{R}_T N - \dot{N} R_T}{N^2}\right] /. \{S \rightarrow s N, I \rightarrow i N, RA \rightarrow r_A N, RT \rightarrow r_T N\}\right]$$

Out[40]=

$$r_A \omega_A - r_T (b + \omega_T)$$

$$\text{In[41]:= } \text{FullSimplify}\left[\dot{s} + \dot{i} + \dot{r}_A + \dot{r}_T\right] /. \{r_A + r_T + s + i \rightarrow 1\}$$

Out[41]=

$$0$$

Check Proportion (COV) Solution Equals Count Solution

In[42]:=

Sol2 =

```

NDSolve[{s'[t] == b - b s[t] - s[t] β ι[t] × N[t] + rT[t] ω_T, ι'[t] == -ι[t] (b + γ - s[t] β N[t]),
  rA'[t] == γ ι[t] - rA[t] (b + ω_A), rT'[t] == rA[t] ω_A - rT[t] (b + ω_T),
  N'[t] == b N[t] - N[t] (μ + k N[t]), N[0] == 10000, s[0] == 9999 / N[0],
  ι[0] == 1 / N[0], rA[0] == 0 / N[0], rT[0] == 0 / N[0]} /. Pars,
{s[t], ι[t], rA[t], rT[t], N[t]}, {t, 0, plottime}]

```

Out[42]=

```

{{s[t] → InterpolatingFunction[Domain: {{0, 1.46 × 103}}] [t],
ι[t] → InterpolatingFunction[Domain: {{0, 1.46 × 103}}] [t],
rA[t] → InterpolatingFunction[Domain: {{0, 1.46 × 103}}] [t],
rT[t] → InterpolatingFunction[Domain: {{0, 1.46 × 103}}] [t],
N[t] → InterpolatingFunction[Domain: {{0, 1.46 × 103}}] [t]}}

```

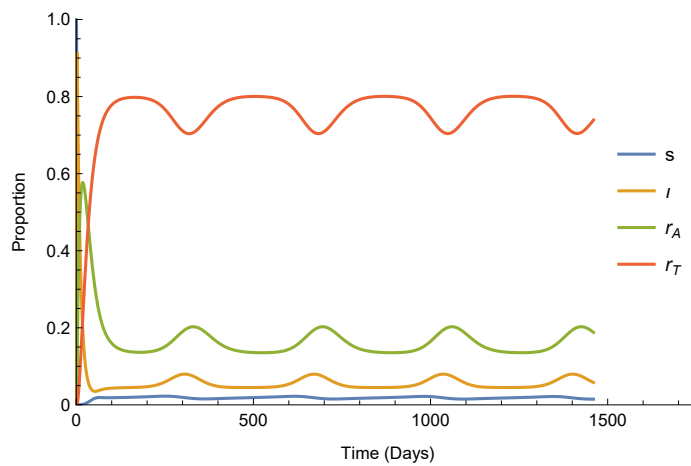
In[43]:=

```

COVPlot = Plot[{s[t] /. Sol2, ι[t] /. Sol2, rA[t] /. Sol2, rT[t] /. Sol2},
{t, 0, plottime}, PlotRange → {{0, plottime + 300}, {0, 1}},
PlotLegends → Placed[{"s", "ι", "r_A", "r_T"}, {Right, Center}],
Axes → False, Frame → {{True, False}, {True, False}},
FrameLabel → {"Proportion", None}, {"Time (Days)", None}, FrameTicks → All]

```

Out[43]=



```
In[44]:= CountSol2 = NDSolve[{S'[t] == b N[t] -  $\beta$  S[t]  $\times$  I[t] - S[t] ( $\mu$  + k N[t]) +  $\omega_T$  RT[t], I'[t] ==
 $\beta$  S[t]  $\times$  I[t] -  $\gamma$  I[t] - I[t] ( $\mu$  + k N[t]), RA'[t] ==  $\gamma$  I[t] - RA[t] ( $\mu$  + k N[t]) -  $\omega_A$  RA[t],
RT'[t] ==  $\omega_A$  RA[t] - RT[t] ( $\mu$  + k N[t]) -  $\omega_T$  RT[t], N'[t] == b N[t] - N[t] ( $\mu$  + k N[t]),
N[0] == 10000, S[0] == 9999, I[0] == 1, RA[0] == 0, RT[0] == 0} /. Pars,
{S[t], I[t], RA[t], RT[t], N[t]}, {t, 0, plottime}]
```

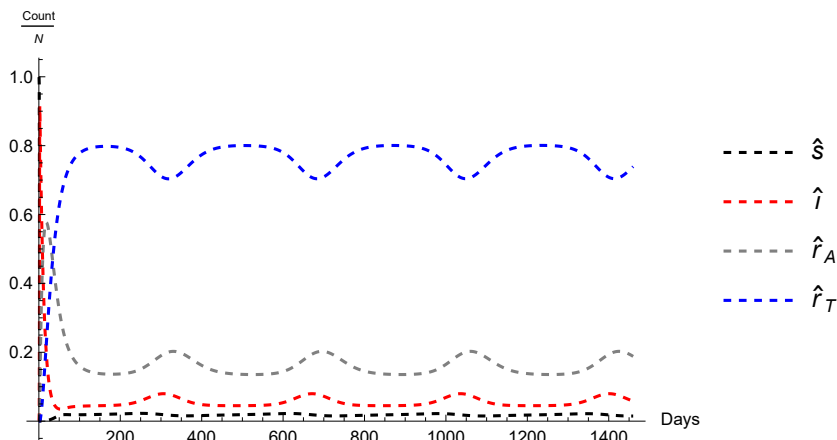
```
Out[44]=
```

```
{ {S[t] → InterpolatingFunction[ Domain: {{0, 1.46 × 10³}}] [t],
I[t] → InterpolatingFunction[ Domain: {{0, 1.46 × 10³}}] [t],
RA[t] → InterpolatingFunction[ Domain: {{0, 1.46 × 10³}}] [t],
RT[t] → InterpolatingFunction[ Domain: {{0, 1.46 × 10³}}] [t],
N[t] → InterpolatingFunction[ Domain: {{0, 1.46 × 10³}}] [t]] }
```

```
In[45]:= CountPlot =
```

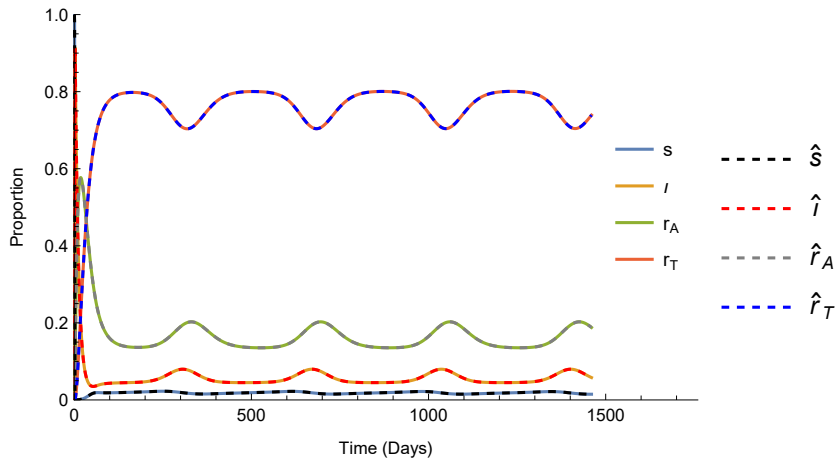
```
Plot[ {  $\frac{S[t]}{S[t] + I[t] + RA[t] + RT[t]}$  /. CountSol2,  $\frac{I[t]}{S[t] + I[t] + RA[t] + RT[t]}$  /. CountSol2,
 $\frac{RA[t]}{S[t] + I[t] + RA[t] + RT[t]}$  /. CountSol2,  $\frac{RT[t]}{S[t] + I[t] + RA[t] + RT[t]}$  /. CountSol2 },
{t, 0, plottime}, PlotLegends → {" $\hat{S}$ ", " $\hat{I}$ ", " $\hat{R}_A$ ", " $\hat{R}_T$ "}, AxesLabel → {Days, Count / N},
PlotStyle → {{Black, Dashed}, {Red, Dashed}, {Gray, Dashed}, {Blue, Dashed}} ]
```

```
Out[45]=
```



In[46]:= Show[{COVPlot, CountPlot}]

Out[46]=



$$\hat{I}(t) = \frac{r_A(b + \omega_A) + \dot{r}_A}{\gamma}$$

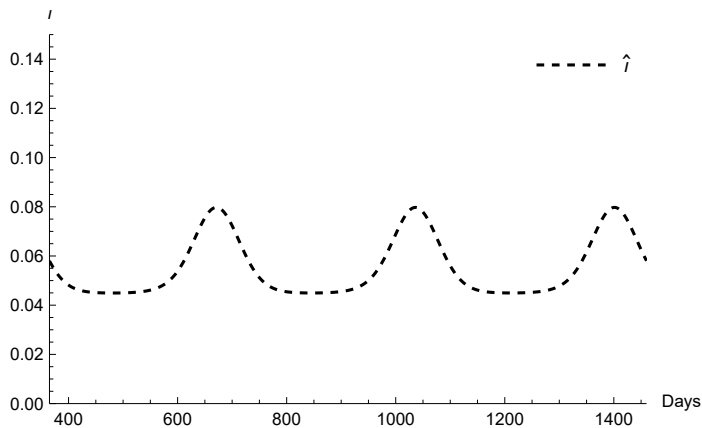
In[47]:= IPredProp = $\frac{D[r_A[t] /. \text{Sol2}, t] + (r_A[t] /. \text{Sol2}) (b + \omega_A)}{\gamma}$ /. Pars

Out[47]=

$$\left\{ 10 \left(\frac{1}{30} + 0.00444145 e^{-4 \cos \left[1 - \frac{\pi t}{365} \right]^2} \right) \text{InterpolatingFunction} \left[\left\{ \left\{ 0., 1.46 \times 10^3 \right\} \right\} \right] [t] + \right. \\ \left. \text{InterpolatingFunction} \left[\left\{ \left\{ 0., 1.46 \times 10^3 \right\} \right\} \right] [t] \right\}$$

In[48]:= PredProp = Plot[{IPredProp}, {t, 365, plottime},
PlotStyle -> {Black, Dashed}, PlotRange -> {{365, plottime}, {0, 0.15}},
AxesLabel -> {Days, I}, PlotLegends -> Placed[{"I-hat"}, {Right, Top}]]

Out[48]=

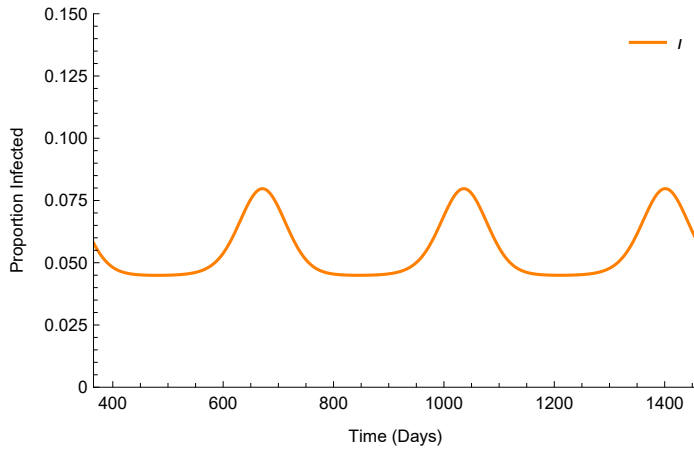


```

In[49]:= TRUEProp =
Plot[{ $\iota[t]$  /. Sol2}, {t, 365, plottime}, PlotRange → {{365, plottime}, {0, 0.15}},
PlotStyle → {Orange}, Axes → False, Frame → {{True, False}, {True, False}},
FrameLabel → {{ "Proportion Infected", None}, {"Time (Days)", None}},
FrameTicks → All, PlotLegends → Placed[{" $\iota$ "}, {Right, Top}]

```

Out[49]=

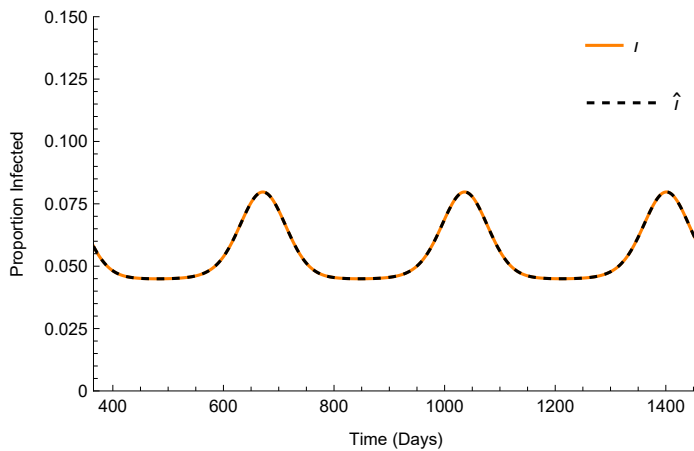


```

In[50]:= TrueBirth = Show[TRUEProp, PredProp]

```

Out[50]=



The Solution for $\hat{I}(t)$ can be approximated without the need for $b(t)$ as long as the birth rate at any time point (t) is $\ll \omega_a$.

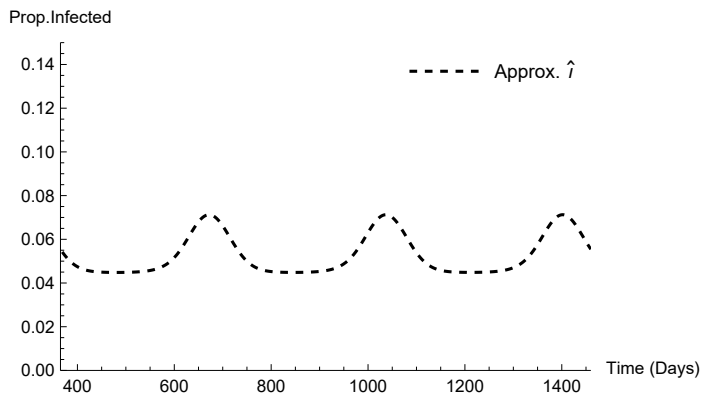
```
In[51]:= IPredPropapprox =  $\frac{D[rA[t] /. Sol2, t] + (rA[t] /. Sol2) \omega_A}{\gamma}$  /. Pars
```

```
Out[51]=
```

$$\left\{ 10 \left(\frac{1}{30} \text{InterpolatingFunction}\left[\left[\begin{array}{c} \text{Domain: } \{ \{0., 1.46 \times 10^3\} \} \\ \text{Output: scalar} \end{array} \right] [t] + \right. \right. \right. \\ \left. \left. \left. \text{InterpolatingFunction}\left[\left[\begin{array}{c} \text{Domain: } \{ \{0., 1.46 \times 10^3\} \} \\ \text{Output: scalar} \end{array} \right] [t] \right] \right\} \right\}$$

```
In[52]:= PredPropapprox = Plot[{IPredPropapprox}, {t, 0, plottime}, PlotStyle -> {Black, Dashed},
  PlotRange -> {{365, plottime}, {0, 0.15}}, AxesLabel -> {"Time (Days)", "Prop. Infected"},
  PlotLegends -> Placed[{"Approx.  $\hat{I}$ "}, {Right, Top}]]
```

```
Out[52]=
```



```
In[53]:= NoBirth = Show[TRUEProp, PredPropapprox]
```

```
Out[53]=
```

